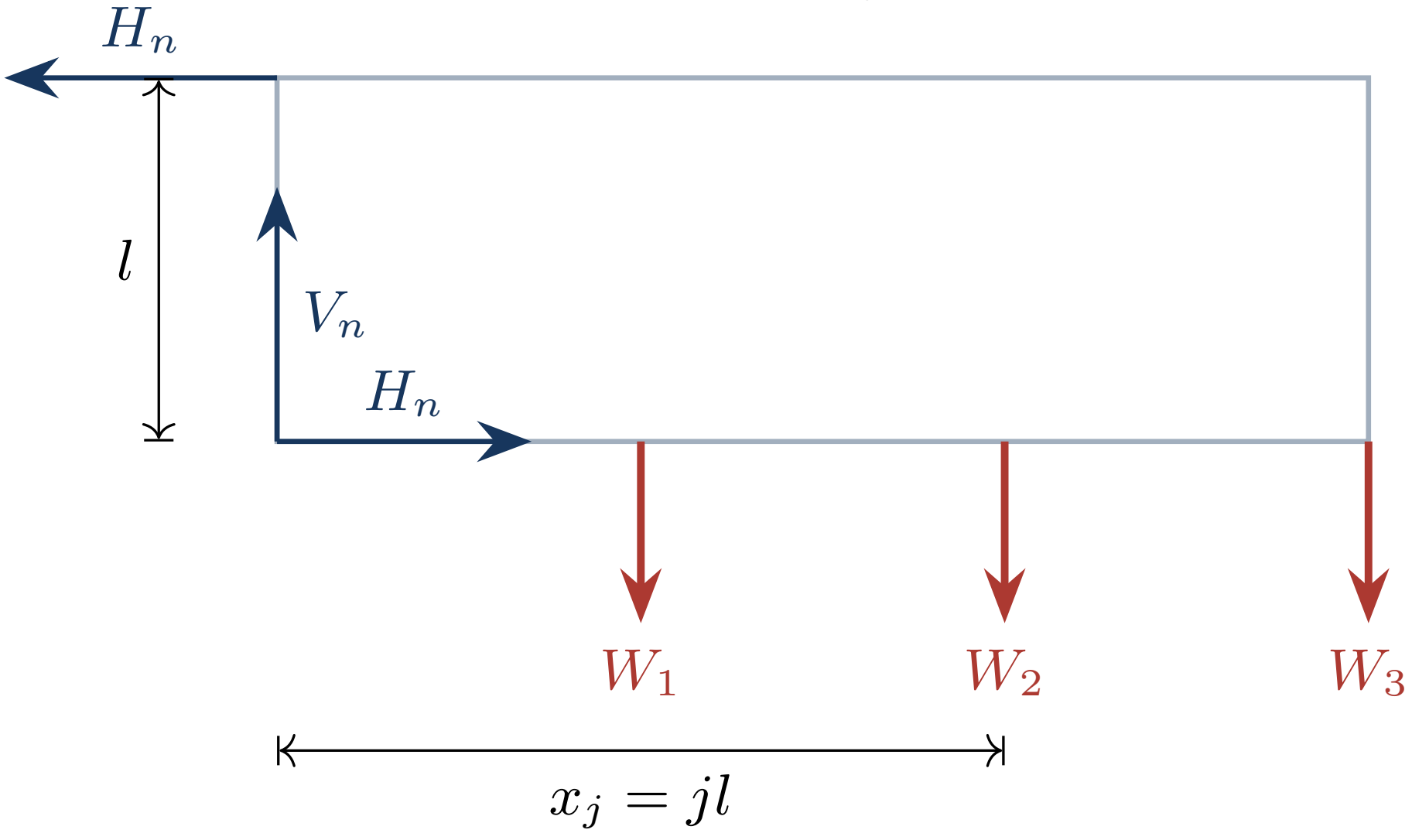
$$H_n l = \sum_j W_j j l$$



上支座竖直反力为 0；下支座承担全部竖直载荷